



The Cell

Key Concepts

- Plant cell: Cell organelles and their functions
- Animal cell: Cell organelles and their functions
- Diagrams of plant and animal cell
- Only the following to be included: Cell wall, Cell membrane, Plastids, Nucleus, Vacuole, Cytoplasm – their structure and functions
- Differences between plant and animal cells

CELL – THE BASIC UNIT OF LIFE

The world is made up of living and non-living things. The body of all the living things is made up of very small, microscopic unit called **cell**. *Cell is the structural and functional unit of life.* A cell undergoes repeated division to give rise to an organism. Each cell performs a specialized function in the body. Like, the cells of the retina of human eye convert the light striking on it into electric signals that are relayed to the brain. These signals are then translated by brain into the images that a person sees. Therefore, any function performed by the body is due to the activity of cells of body. Thus, **cells** are also called **building blocks of life**.

DISCOVERY OF THE CELL *would be tested in exam*

The cell is very minute and thus was not discovered before the invention of microscope. *Microscope is a device used to see objects not visible to naked eyes.* It was Antonie van Leeuwenhoek (1632–1723), who first developed a simple microscope which is made up of a single biconvex lens. He used it to see the blood cells in the capillaries of a frog's foot. He identified them as living units of living beings.

Robert Hooke (1635–1703), an English scientist, coined the term "Cell" to explain the basic unit of life. In 1660, Hooke developed a compound microscope which was made up of two lenses. He used this microscope to see a thin slice of a dead cork. In this slice, he observed some "honeycomb" like structures and named them "cells" (Fig. 3.1).

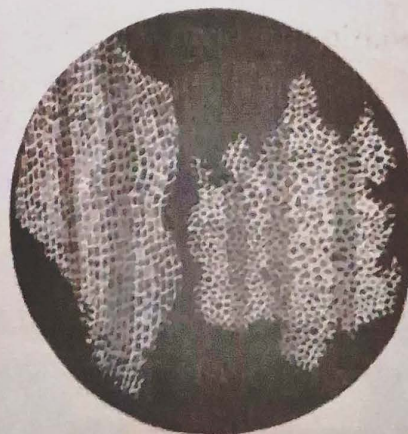


Fig. 3.1: Honeycomb like structure of cell

The word cell is derived from a Latin word, *cellula*, which means small compartment. Thus, it was Robert Hooke who first discovered and named cell.

CELL THEORY

Much advancement in development of microscopes took place in the next two centuries. In 1838, German scientist, Matthias Jakob Schleiden used microscope to observe different parts of a plant. He found that these parts of the plant were made up of cells. Another, German scientist, Theodor Schwann used microscope to study different parts of animals and found that these too were made up of cells. An important statement made by him was "all life processes take place inside cells". After fifty years, a German scientist, Rudolf Carl Virchow, discovered that a cell divides to form new cells.

On the basis of these remarkable findings, the basic features of modern cell theory states that

1. All living things are made up of one or more cells.
2. Cells are the basic structural and functional units of all living beings.
3. All cells are basically the same in chemical composition and metabolic processes.
4. All new cells arise by the division of pre-existing cells.

NUMBER, SHAPE AND SIZE OF THE CELL

Cells differ greatly in their shape (Fig. 3.2), size, and number. Some living organisms such as *Amoeba* are composed of one cell and are called **unicellular organisms**. Large organisms such as elephant, human, tree, etc. are composed of millions of cells and are called **multicellular organisms**. Table 3.1 shows the diversity in cells in terms of number, shape, and size.

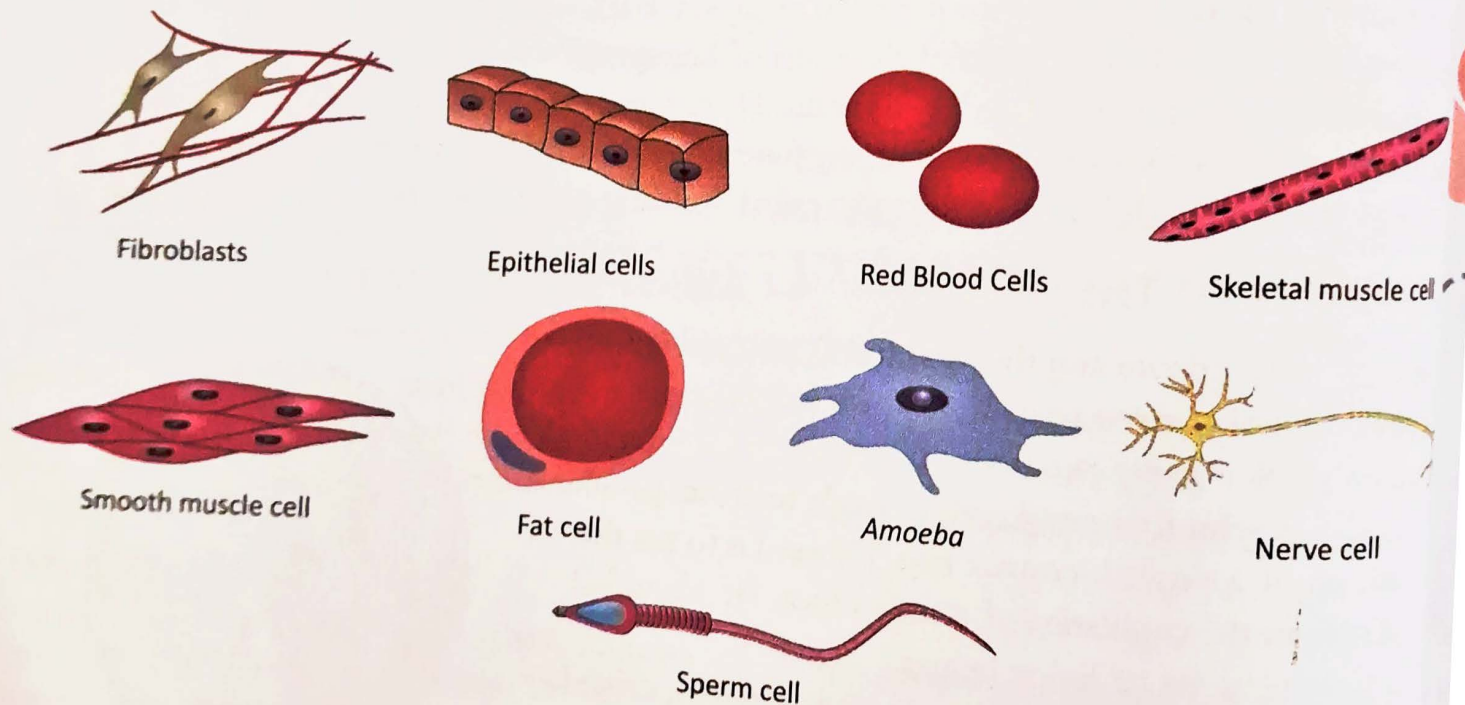


Fig. 3.2: Different shapes of cells

Did you know

The branch of biology that deals with the study of cells is called cytology.

★ **Table 3.1: Number, shape and size of cells**

Number of cells	One-celled or Unicellular: Organisms with one-celled body such as <i>Amoeba</i>, <i>Chlamydomonas</i>, <i>Paramecium</i> Multi-celled or Multicellular: Organisms with many cells in the body such as elephant, rose, tiger, human, insects, birds, etc.
Shape of cells	Irregular: They lack fixed shape. For example, <i>Amoeba</i> Oval: Organisms with oval-shaped cell. For example, <i>Chlamydomonas</i> Oblong: Organisms made of oblong cell such as <i>Paramecium</i> Elongated: These are elongated, contractile cells such as muscle cells Thread-like: Cells are thread-like. For example, nerve cells
Size of cells	Mostly cells are microscopic. An average cell size ranges from 3 to 30 μ. In human body, red blood cells are the smallest. They are 0.1 μ in diameter. The bacterial cell is the smallest cell which measures about 0.2 μ to 0.5 μ. The length of an ostrich's egg is about 23 cm and elephant's nerve cell is the longest cell, i.e. 3 m.

CELL – THE STRUCTURAL UNIT OF LIFE

As we have studied, cells vary in their number, shape, and size. However, the basic structural components are same. Every part of plant and animal's body is made up of cells. If you observe a small portion of a leaf, root tip, stem, etc. they are made up of cells.

Did you know

Cell size is measured in a unit called micron (μ). One micron is equal to 1/1000th of a millimetre.

Aim: To observe cells in an onion peel

Materials Required: A small portion of onion leaf, forceps, dilute iodine solution, glass slide, coverslip and microscope

Procedure:

1. Take a small portion of a fleshy onion leaf. With the help of the forceps, pull out a thin lining from its inner surface.
2. Cut a small square of this peeled leaf.
3. Place it on a glass slide. Add a drop of dilute iodine solution to it with the help of a dropper.
4. Gently put the coverslip on it. Ensure to avoid the formation of air bubbles inside the coverslip. Observe the slide under the microscope.

Observations: Rectangular-shaped cells are observed (Fig. 3.3).

Inference: The fleshy onion leaves are made up of cells.

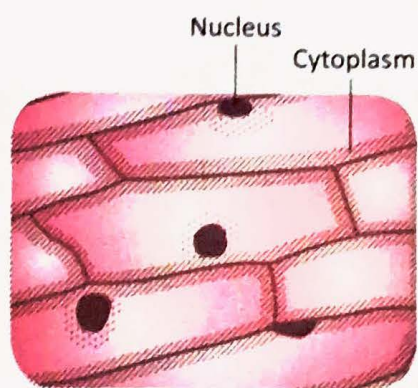


Fig. 3.3: Onion peel cells

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ACTIVITY

Aim: To observe cells in the inner surface of human cheek

Materials Required: A blunt toothpick, methylene blue solution, glass slide, coverslip and microscope

Procedure:

1. Take a blunt toothpick. Gently scrape the inner side of your cheek.
2. Transfer these scrapings onto the glass slide and add a drop of methylene blue to it.
3. Gently put the coverslip on it and observe it under the microscope.

Observations: Cheek cells are present in the smear from cheeks (Fig. 3.4).

Inference: The human cheeks are made up of cells.

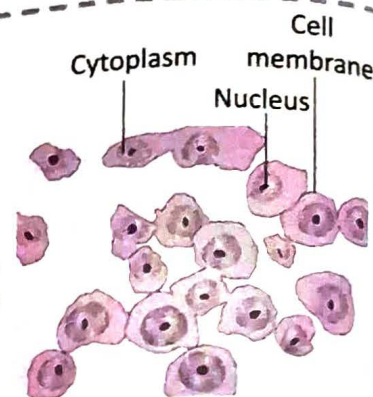


Fig. 3.4: Human cheek cells

CELL – THE FUNCTIONAL UNIT OF LIFE

All the life processes performed by our body organs are due to the activity of cells present in our body. Table 3.2 shows some examples in plants and animals where cell acts as the functional unit of our body.



Table 3.2: Cell – The functional unit of life in plants and animals

Plants/ Animals	Function	Activity of cell
Plants	Absorption of water and minerals	Root cells of plants absorb water and minerals from soil. If root dies, the plant will also dry up.
	Preparation of food	Green pigment (chlorophyll) containing cells utilize the energy of the sun, carbon dioxide and water to prepare food for plants (photosynthesis).
Animals	Movement	Contraction and expansion of muscle cells help to move the body or fold and unfold the body parts.
	Digestion	Cells of various glands secrete digestive enzymes that help to digest the food
	Release of energy	All the energy needed by our body to perform various activities (inside the body of an organism) is derived from the oxidation of food inside the cells (respiration)

CheckPoint



Fill in the blanks.

1. _____ is the basic unit of life.
2. All the _____ performed by our body organs are due to the _____ of cells present in our body.
3. Every part of _____ and animal's body is made up of _____.
4. The bacterial cell is the _____ cell which measures about 0.2μ to 0.5μ .

BASIC STRUCTURE OF THE CELL

We see diversity in cells in terms of **number, shape and size of cells**. However, the basic structure of a cell remains same. The prime components of a cell are **cell membrane, nucleus, cytoplasm, and cell organelles**.

Cell Membrane

It forms an **outer boundary** of the cell. This is a **selectively permeable membrane**. It allows only **certain substances** to enter and move out of the cell. It is known as **plasma membrane** in plant cells. The plasma membrane is enclosed by an outermost protective layer called **cell wall**. The **cell membrane is elastic** and thus provides **stretch ability** to the cell to **some extent**.

Cell Wall

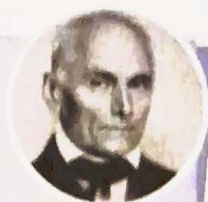
Only plant and bacterial cells possess cell wall. It is an **outermost protective layer** of cell. It is a **non-living structure**. The cell wall is made up of **cellulose**. It gives **shape and rigidity** to the cell. Cell wall **allows substances in a solution to enter and leave the cell freely**. Thus, it is a **highly permeable layer**.

★ **Table 3.3:** Differences between cell wall and cell membrane ★

Cell wall	Cell membrane
1. It is an outer protective layer of cell.	1. It is the outer covering of the protoplasm of cell.
2. It is a non-living structure.	2. It is a living structure.
3. It is freely permeable layer.	3. It is selectively permeable layer.
4. It provides rigidity and shape to the cell.	4. It provides stretch ability to the cell.
5. It is found only in plant and bacterial cells.	5. It is found in both animal and plant cells.

Science
Guru

J E Purkinje (1787–1869) was a Czech physiologist. One of his best known discoveries is of large nerve cells with many branch extensions found in the cortex of the cerebellum of brain. While studying the young animal embryos, he coined the term protoplasm. Protoplasm is colourless, jelly like viscous substance. The protoplasm which is present in the nucleus (denser part) is called nucleoplasm. The protoplasm which surrounds the nucleus is called cytoplasm (semi-solid jelly like part).



Nucleus

Nucleus is a **small spherical mass** present in the **centre** of the **cytoplasm** (Fig. 3.5). It has a **surrounding membrane** called nuclear membrane. The nuclear membrane is filled with **dense nucleoplasm**. It contains certain thread-like structures called **chromatin fibres**. During cell

division, these fibres become thick and are called **chromosomes**. *Chromosomes carry genes which contain hereditary information of the organism.* In humans, there are **23 pairs** of chromosomes in each cell. The number of chromosomes is **fixed** in every species.

Functions: Nucleus regulates and coordinates various life activities in a cell. It plays an important role in cell division. It also contains hereditary information of an organism in genes.

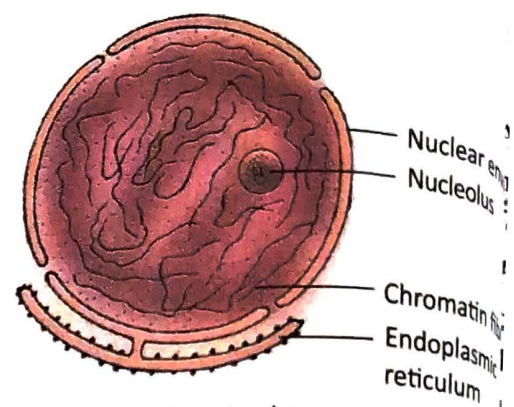


Fig. 3.5: Nucleus

Cytoplasm

It is a semi-liquid substance of a cell which is enclosed by cell membrane. With the help of the much advanced microscope, the electron microscope, it was observed that many tiny structures are embedded in the cytoplasm. These are called **cell organelles**. They are not visible by a compound microscope. Many cellular processes take place within cytoplasm with help of these cell organelles. These structures produce many substances such as enzymes, hormones, proteins, etc.

Cell Organelles

The following are some of the main cell organelles present in the cytoplasm:

Endoplasmic Reticulum

Endoplasmic reticulum is an irregular network of fine tubules. These tubular structures are bounded by membrane and are interconnected with each other. Many times, these tubules join the nuclear membrane at one end and to the plasma membrane on the other end. Depending on the absence or presence of ribosomes on the surface of endoplasmic reticulum, they can be smooth endoplasmic reticulum (those without ribosomes on surface) or rough endoplasmic reticulum (those with ribosomes on the surface) (Fig. 3.6).

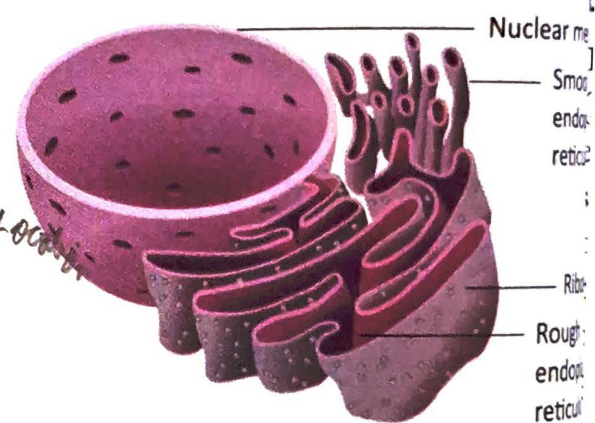


Fig. 3.6: Endoplasmic reticulum

★ **Functions:** It serves as a pathway for the distribution of materials within a cell and between two cells. It also provides internal support to the cell.

Ribosomes

They are very minute structures scattered in cytoplasm or also attached to the surface of endoplasmic reticulum. They do not possess membrane.

★ **Function:** They help in the synthesis of proteins.

Mitochondria

Mitochondria (*Sing.* Mitochondrion) occur in the form of small and spherical rod-shaped structures. They range from few hundreds to a few lakhs in number in the cytoplasm of a single

A mitochondrion consists of an outer membrane and an inner membrane (Fig. 3.7). The inner membrane is folded inward to form a finger-like structure called cristae.

Functions: They help in cellular respiration. They oxidize food by utilizing oxygen and release energy in the form of Adenosine Tri Phosphate (ATP). Thus, they are also known as powerhouses of the cell.

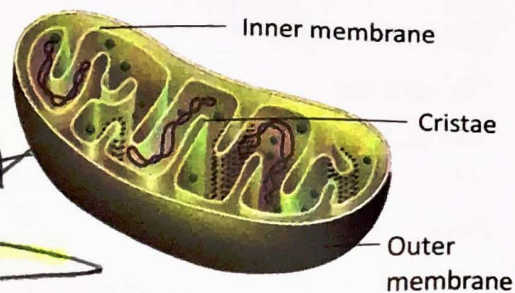


Fig. 3.7: Mitochondrion

Golgi Bodies

These are the membrane bound stacks of thin, flattened, and parallel pouches called cisternae (Fig. 3.8). These cisternae are not connected to each other. They are found in cytoplasm near the nucleus that is next to the endoplasmic reticulum. In plant cells, golgi bodies are smaller and are more in number. They are also called dictyosomes. Golgi bodies are absent in bacteria, certain algae and mature sperm cells and red blood cells.

Functions: Golgi bodies help in secretion of certain hormones, enzymes, etc. They help in the formation of cell wall in plant cells. They are also involved in transport of substances within the cell.

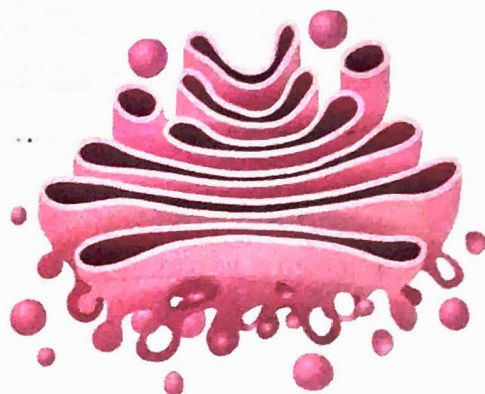


Fig. 3.8: Golgi bodies

Lysosomes

These are single membrane bound small sac like structures (Fig. 3.9). They are present in cytoplasm. They contain enzymes that help in the digestion of various materials. Under some circumstances, lysosomes release enzymes that digest other cell organelles. Thus, they are also known as suicide bags of the cell. Mostly lysosomes are found in animal cell but may also be present in plant cell.

Functions: Lysosomes help in the digestion of food within the cell. They protect the body from bacteria and other foreign bodies such as germs that enter the cell. They also help in the removal of old worn-out cell organelles.

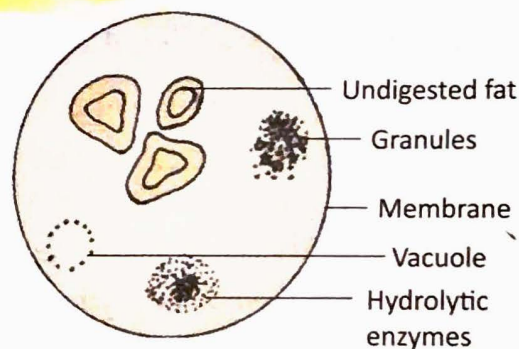


Fig. 3.9: Lysosome

Centrosome - Animal cell

A centrosome is an oval or rounded non-membranous organelle. This cell organelle is found in animal cells only. It is present near the cell nucleus in the cytoplasm. It consists of two minute granules called centrioles (Fig. 3.10).

Function: It initiates and regulates cell division in animal cells.

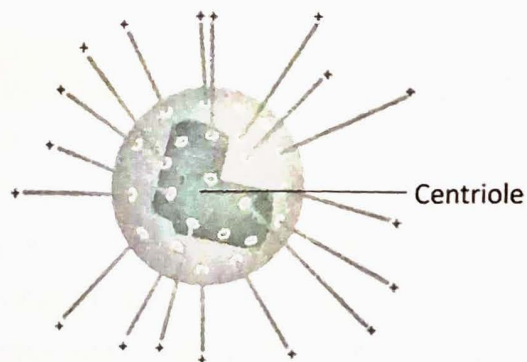


Fig. 3.10: Centrosome

Plastids

Plastids are found in plant cells only. These are found in cytoplasm in the form of oval, spherical or disc-shaped structure. The organelle manufactures and stores pigments. There are three types of plastids—chloroplasts, chromoplasts, leucoplasts.

Function

1. **Chloroplasts:** Chloroplasts contain green-coloured pigment called **chlorophyll**. They are also known as **kitchen of the cell** as they trap the solar energy of the sun and convert it into chemical energy of **food for plants** during photosynthesis. They are responsible for imparting **green colour to leaves and other parts of the plant where they are found**. Structurally, chloroplast consists of two regions, the grana and the stroma (Fig. 3.11).
2. **Chromoplasts:** Chromoplasts contain coloured pigments except green. They are found in fruits and petals of flowers. They are responsible for imparting colours such as **red, yellow or orange** (Fig. 3.12).
3. **Leucoplasts:** Leucoplasts lack any pigment, so they are colourless plastids. They help in storing food prepared by plants in the form of **starch, fats and proteins**.

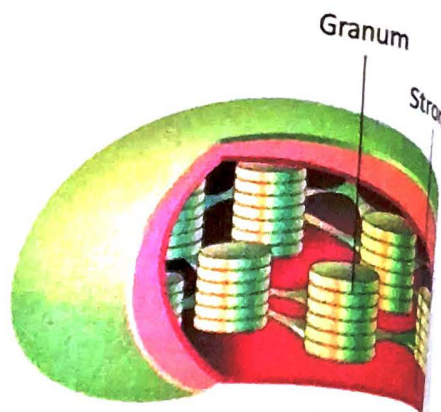


Fig. 3.11: Chloroplast



Fig. 3.12: Chromoplast found in fruits and flowers

Vacuoles - *Star food and water*

Vacuoles are fluid-filled cavities in the cytoplasm. They also contain water. They are covered by a membrane called tonoplast. Plant cells have more and larger-sized vacuoles as compared to animals. However, some plant cells have only one large vacuole. Animal cells may or may not have vacuoles.

Functions: Vacuoles help in storage of food, water and other substances. They also help in maintaining the shape of cell by making it rigid. They help in excretion.

Check Point



Write True or False for the following sentences.

1. Chromosomes carry genes which contain hereditary information of an organism.
2. Golgi bodies are known as powerhouse of a cell.
3. Ribosomes are membranous structure present on the endoplasmic reticulum.
4. Centrioles help in regulating cell division.
5. Leucoplasts are plastids that contain red-coloured pigment.

DIFFERENCES BETWEEN PLANT AND ANIMAL CELLS

Table 3.4 shows some differences found between plant and animal cells (Fig. 3.13). However, the basic structure of a cell remains the same.

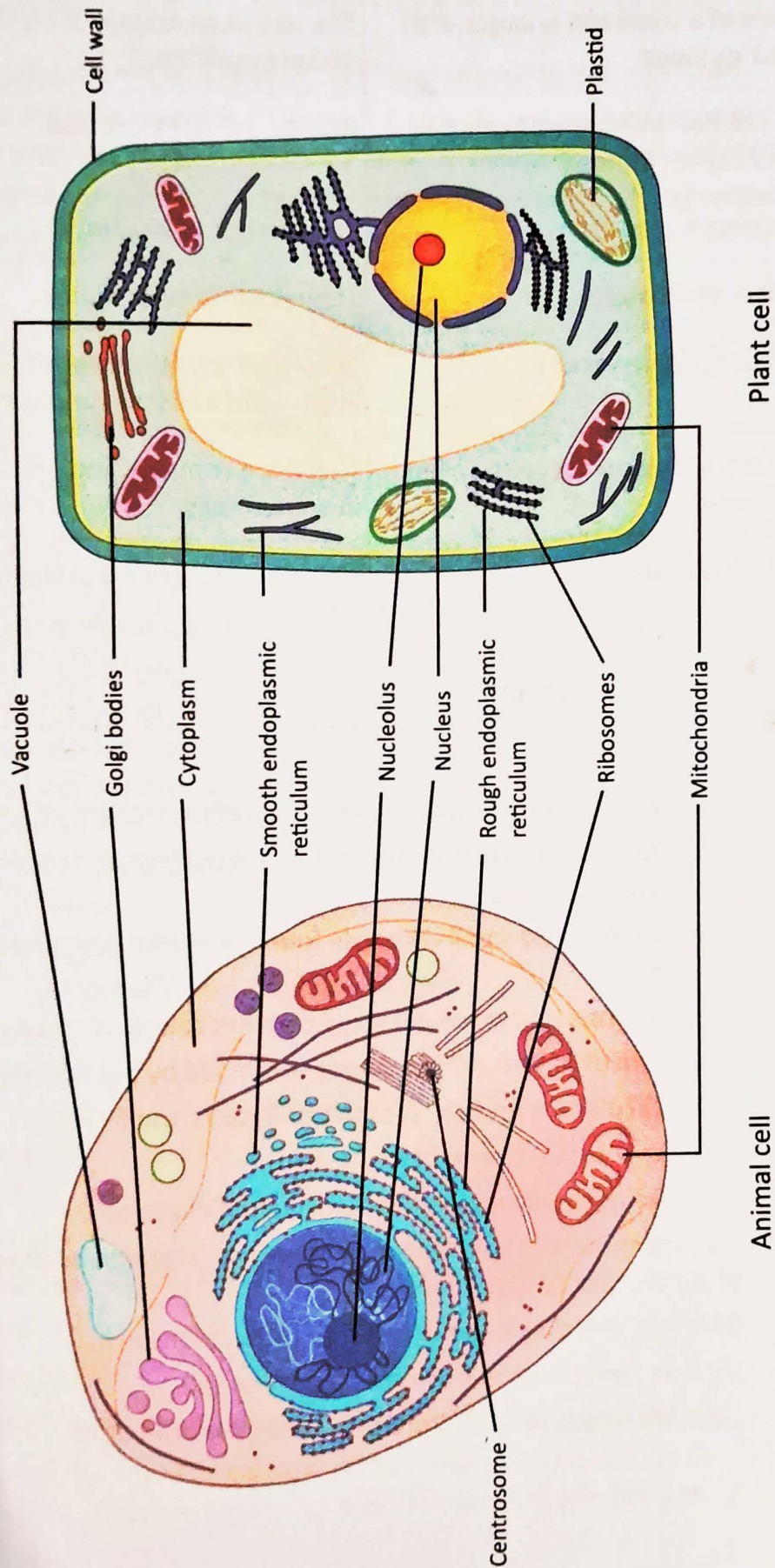


Fig. 3.13: Structure of plant and animal cells

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Table 3.4: Differences between plant cell and animal cell

Characteristic	Plant cell	Animal cell
Cell size	The size of a plant cell is larger with distinct outlines.	The size of an animal cell is smaller with distinct outlines.
Cell wall	Plant cell has well-defined cell wall around cell membrane.	Animal cell lacks cell wall.
Plastids	They contain plastids.	They do not have plastids.
Centrosome	They lack centrosome.	They possess centrosome.
Vacuole	They have a large vacuole.	They have small temporary vacuoles. They are involved in excretion and secretion.
Golgi bodies	These are simpler units, also called dictyosomes.	These are complex and prominent structures in animal cells.
Lysosomes	They lack lysosomes.	They have lysosomes in the cytoplasm.

Keywords



Cell wall:	Protective covering present outside the cell membrane of plant cells
Chromosome:	Small, thread-like structure present in the nucleus and carries hereditary information
Centrosome:	Non-membranous small organelle found in animal cells and involved in division
Chloroplast:	Plastids containing green-coloured pigment called chlorophyll
Chromatin:	A network of thread-like structures contained by the nucleoplasm
Chromoplast:	Colourful plastids present in petals of flowers and fruits
Cytoplasm:	Jelly-like substance present in the cell
Endoplasmic reticulum:	Tube-like organelle involved in transport of material
Leucoplast:	Colourless plastids that help in storing food prepared by the plant in the form of starch, proteins, and fats
Mitochondrion:	Cell organelle that is the site of respiration in the cell
Plasma membrane:	The cell membrane enclosing all the living cells
Vacuole:	Liquid-filled sac like structure that keeps the cell turgid